Themistoklis Nikas

Ph.D. Student in Computer Science
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Education

Boston University — Boston, MA, USA
Ph.D. in Computer Science; Advisor: Prof. Aaron Mueller — *Sep 2023 – May 2028 (expected)*
- Mechanistic interpretability of LLMs, focusing on uncertainty, confidence estimation, and calibration. Current work uses linear representations of ambiguity to improve calibration via fine-tuning.

University of Ioannina — Ioannina, Greece
B.Sc. & M.Sc. in Computer Science & Engineering (5-year joint degree; 300 ECTS) — *Mar 2017 – Mar 2022*
- Diploma Thesis: *Fairness in Link Analysis Algorithms*: achieving fairness in PageRank with minimal network modifications, with theoretical and empirical analysis; led to a publication.

Publications

Improving Confidence Estimation in Language Models with Linear Representations of Ambiguity
***Nikas, T.**, Subramani, N., Mueller, A. (Actionable Interpretability @ COLM 2026); under submission*

Minimizing the Cost of PageRank Fairness
*Tzimas, S., **Nikas, T.**, Pitoura, E., Tsaparas, P. (WebGraph @ WSDM 2026)*

Disentangling Text and Math in Word Problems: Evidence for the Bidimensional Structure of Large Language Models' Reasoning
*Calais, P., Franco, G., Tang, Z., **Nikas, T.**, Meira Jr., W., Terzi, E., Crovella, M. (Findings of ACL 2025)*

Beyond Accuracy: Understanding the Performance of LLMs on Exams Designed for Humans
*Calais, P., Franco, G., **Nikas, T.**, Tang, Z., Crovella, M., Meira Jr., W., Terzi, E. (Preprint, 2025)*

Research Experience

Research Associate — Feb 2023 – Aug 2023
University of Ioannina & Pfizer CDI Collaboration — *Thessaloniki, Greece*
- Built ML/NLP tools to assess health misinformation on Reddit, collecting subreddits on hoaxes and rumors and verifying claims with fact-checking tools and databases.
- Clustered posts with text embeddings and analyzed them via graph analysis, feature extraction, and statistics.

Junior Researcher — May 2022 – Jan 2023
Unit of Medical Technology & Intelligent Information Systems (MedLab) — *Ioannina, Greece*
- Researched probabilistic and ML methods for synthetic omics generation and virtual population analysis.
- Investigated the relationship between cardiovascular diseases and depression using ML classification.

Teaching Fellow

Boston University — Jan 2024 – Dec 2025
- Introduction to Analysis of Algorithms (Spring 2024, Fall 2024, Spring 2025, Fall 2025). Led discussion sections, held office hours, and helped design assignments and exams.

University of Ioannina — Feb 2023 – Jun 2023
- Object-Oriented Programming; supported students through labs, grading, and Q&A.

Service & Activities

Reviewer: Actionable Interpretability @ COLM 2026
Co-organizer: New England Mechanistic Interpretability Workshop (NEMI 2026)
Talk: *Minimizing the Cost of PageRank Fairness*, WebGraph @ WSDM 2026

Skills, Languages & Volunteering

Programming & Tools: Python, PyTorch, Hugging Face, TransformerLens, nnsight, scikit-learn, Git, LaTeX
Languages: English (fluent), Greek (native)
Volunteering (TEDxMaviliSquare, 2021–2023): Oversaw speaker recruitment, content direction, and workshops.